

1. Administrative & Study Identification

Full Study Title: A Phase 2 Study of Pembrolizumab (MK-3475) Every 6 Weeks (Q6W) in Participants With Relapsed or Refractory Classical Hodgkin's Lymphoma (rrcHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL) **Short Title/Acronym:** KEYNOTE-B68 (MK-3475-B68) **Protocol Number:** MK-3475-B68 **NCT Number:** NCT04875195 **Posting ID:** 628098404317821277 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475) / Trade Name: Keytruda **Phase of Development:** Phase 2 **Trial Status:** Completed **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the objective response rate (ORR), by cohort (rrcHL and rrPMBCL), as assessed by the investigator according to Lugano classification criteria 2014 in participants treated with pembrolizumab every six weeks (Q6W). **Secondary Objectives:** To evaluate the ORR per Blinded Independent Central Review (BICR), duration of response (DOR) per investigator and per BICR, progression-free survival (PFS), overall survival (OS), and safety profile of the treatment.

2.2 Background & Rationale Pembrolizumab 200 mg every 3 weeks (Q3W) is approved for the treatment of relapsed/refractory classical Hodgkin lymphoma (cHL) and primary mediastinal B-cell lymphoma (PMBCL). The FDA granted accelerated approval to pembrolizumab 400 mg Q6W based on pharmacokinetic data in solid tumors. This KEYNOTE-B68 study was initiated to prospectively evaluate the efficacy and safety of the pembrolizumab 400 mg Q6W regimen in patients with R/R cHL or R/R PMBCL, aiming to provide patients with a more convenient treatment schedule while maintaining durable disease control.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Single-Group Assignment) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 66 patients (approx. 60 R/R cHL and 6 R/R PMBCL) **Number of Sites:** Global, Multicenter **Estimated Study Start:** 07-Jun-2021 **Study Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. ≥ 18 years old with a histologically confirmed diagnosis of cHL or PMBCL according to the World Health Organization (WHO) classification. 2. Has radiographically measurable cHL or PMBCL disease per Lugano classification with at least 1 nodal lesion (not previously radiated) >15 mm in long axis, and/or an extranodal lesion >10 mm in long and short axis. 3. **rrcHL cohort:** Have relapsed during their last cHL regimen after receiving at least 2 cycles of therapy or within 12 months after completing the last regimen; OR have received ≥ 1 line of prior multiagent therapy (with/without brentuximab vedotin) or auto-SCT and have failed to respond or relapsed. 4. **rrPMBCL cohort:** Have relapsed after auto-SCT or failed to achieve a CR or PR within 60 days of auto-SCT; OR for participants ineligible for auto-SCT, have received ≥ 2 prior lines of therapy (including at least 1 rituximab-based regimen) and failed to respond or relapsed. Participants should not need urgent cytoreductive therapy. 5. Must submit an evaluable core lymph node biopsy for biomarker analysis from an archival (>60 days) or newly obtained (within 30 days) sample which was not previously irradiated. If no archival tissue is available, 2 new fresh core needle samples are required. 6. Female participants must not be pregnant or breastfeeding. Women of childbearing potential (WOCBP) must use highly effective contraception (failure rate $<1\%$ per year) or remain abstinent for at least 120 days after the last dose of study intervention. 7. Has an ECOG performance status of 0 to 1, life expectancy >3 months, and adequate organ function.

4.2 Exclusion Criteria 1. Has undergone a solid organ transplant at any time, or prior allogeneic hematopoietic SCT within the last 5 years. 2. Has clinically significant cardiovascular disease: cerebral vascular accident/stroke or myocardial infarction (<6 months prior to enrollment), unstable angina, congestive heart failure (NYHA Class $\geq$ II), or serious cardiac arrhythmia. 3. Has pericardial effusion or clinically significant pleural effusion. 4. Has a history of a second malignancy, unless potentially curative treatment was completed with no evidence of malignancy for 2 years (exceptions include successfully resected basal/squamous cell skin cancer, superficial bladder cancer, or in situ cancers). 5. Has received prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2 agent, or an agent directed to another stimulatory/co-inhibitory T-cell receptor (e.g., CTLA-4, OX-40, CD137), or prior CAR-T therapy. 6. Is receiving systemic steroid therapy (>10 mg daily prednisone equivalent) <3 days prior to the first dose, or any form of immunosuppressive therapy within 7 days prior (except daily replacement therapy). 7. Has received a prior monoclonal antibody, systemic anticancer therapy, or investigational agent within 4 weeks prior to the first dose. 8. Has received prior radiotherapy within 2 weeks of start of study intervention, or has a history of radiation pneumonitis requiring corticosteroids.

9. Has received a live or live-attenuated vaccine within 30 days before the first dose. 10. Has known active central nervous system (CNS) lymphoma involvement. 11. Has severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab and/or its excipients. 12. Has an active autoimmune disease requiring systemic treatment in the past 2 years, or a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids (or current pneumonitis/ILD). 13. Has an active infection requiring systemic therapy, known history of HIV infection, known Hepatitis B (HBsAg reactive), or active Hepatitis C virus infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab 400 mg every 6 weeks (Q6W) **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Up to 18 cycles (approximately 2 years), or until disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms and comorbidities, daily steroid replacement therapy. **Prohibited:** Concurrent treatment with other investigational agents; urgent cytoreductive therapy; live vaccines; systemic immunosuppressive therapies >10 mg daily prednisone equivalent.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy, Imaging, Labs
Baseline	Day 0	First Dose (Pembrolizumab 400 mg), Baseline Safety Assessment
Treatment Cycles	Every 6 Weeks	Q6W Dosing, Safety Labs, AE Monitoring, PK sampling
Interim Scans	Every 12 Weeks	Tumor Imaging (CT/FDG-PET), Efficacy Assessment
End of Study	Up to 2 Years	Final Efficacy Assessment (Cycle 18), Exit Interview, Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), by cohort (rrCHL and rrPMBCL), defined as the percentage of participants who had a complete response (CR) or partial response (PR). **Measurement Method:** Assessed by the investigator according to Lugano classification criteria 2014, evaluated using computed tomography (CT) and metabolic imaging (FDG-PET). - *Complete Response (CR):* Complete metabolic (no/minimal FDG uptake) and radiologic response (target lesions regress to ≤ 1.5 cm in longest transverse diameter) and no new lesions. - *Partial*

Response (PR): Partial metabolic (moderate/high FDG uptake) and radiologic response ($\geq 50\%$ decrease in sum of product diameters for multiple lesions of up to 6 target nodes and extranodal sites, no increase in lesions, and spleen regressed by $>50\%$ in length beyond normal).

7.2 Secondary & Exploratory Endpoints - ORR per Lugano

Classification as assessed by Blinded Independent Central Review (BICR). - Duration of Response (DOR) per Lugano Classification as assessed by Investigator. - Duration of Response (DOR) per Lugano Classification as assessed by BICR. - Progression-Free Survival (PFS). - Overall Survival (OS). - Pharmacokinetics (PK) and Safety/Treatment-Related AEs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during study visits, graded by CTCAE criteria. Specific focus on Immune-Mediated Adverse Events (irAEs). **Serious Adverse Events (SAEs)**: Required to be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC)**: External independent review mechanism utilized for safety oversight per sponsor protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The efficacy analysis (Intent-to-Treat) includes all patients who received at least one dose of the study drug. Safety Set includes all participants exposed to treatment for AE/SAE evaluations. **Sample Size Justification**: 66 patients targeted to sufficiently estimate ORR and confirm historical equivalence to the Q3W regimen. **Handling of Missing Data**: Participants with missing post-baseline assessments are generally considered non-responders for ORR calculations; standard right-censoring used for time-to-event (PFS/OS) analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits for source data verification. **Audit Trail**: Utilizing secure EDC (Electronic Data Capture) systems, eCRF locking, and rigorous data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-3475-B68 **NCT Number:** NCT04875195 **Posting ID:** 628098404317821277 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** A Study of Pembrolizumab (MK-3475) in Relapsed or Refractory Classical Hodgkin's Lymphoma (rrcHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL) (MK-3475-B68) **Official Regulatory Title:** A Phase 2 Study of Pembrolizumab (MK-3475) Every 6 Weeks (Q6W) in Participants With Relapsed or Refractory Classical Hodgkin's Lymphoma (rrcHL) or Relapsed or Refractory Primary Mediastinal Large B-cell Lymphoma (rrPMBCL)

12. Clinical Framework (Disease Specific)

Primary Condition: Hodgkin's Lymphoma, Primary Mediastinal Large B-cell Lymphoma (PMBCL) **Diagnosis Threshold:** Histologically confirmed diagnosis (WHO criteria) that is measurable per Lugano classification. **Medical Methodology:** Immune Checkpoint Inhibition (Anti-PD-1 Therapy) **Optimization Target:** Sustained Objective Response (CR/PR)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Relapsed/Refractory Lymphomas **Education Protocol (HCP):** RECIST/Lugano Criteria training and irAE management protocols **Education Protocol (Patient):** Infusion schedule training and immune-related symptom tracking **Quality Audit Mechanism:** Independent central (BICR) or investigator radiological review for response validation

14. Observation & Follow-Up Schedule

Total Duration: Up to 24 months (18 cycles) **Onsite Visit Cadence:** Every 6 weeks (Q6W) during active treatment phase **Remote Monitoring Frequency:** Inter-cycle as clinically indicated **Checklist Review Interval:** Routine efficacy imaging typically every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				